

- 2 (a)** The movement of ions can result in different actions in plant cells and animal cells.

Table 2.1 lists several ions, the direction of movement of each ion and the cell type concerned, and the action that results from the movement of each ion.

Complete Table 2.1.

Table 2.1

ion	direction of movement of ion and type of cell	action that results from movement of ion
K ⁺		opening of stomata
Na ⁺	into synaptic knob of presynaptic neurone	
Ca ²⁺	into synaptic knob of presynaptic neurone	
.....	from sarcoplasmic reticulum to sarcoplasm of cell of striated muscle	binding to troponin

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- (b)** In the nephron of the kidney, selective reabsorption occurs in the cells of the proximal convoluted tubule. Sodium ions are actively transported out of these cells into the surrounding tissue fluid and pass into the blood.

Outline how the active transport of sodium ions out of the proximal convoluted tubule cells results in the absence of glucose in urine.

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[Total: 8]